

SQL Assessment: Car Rental Management System

Hello and Good Morning!

Today's task is to design and implement a **Car Rental Management System database** in SQL Server.

1. Analyse the Entities

- Review the tables listed below and figure out how they relate to each other.
- Example: A **Customer** can have many **Rentals**, but each Rental belongs to one Customer (One-to-Many).

2. Define Relationships and Cardinalities

- Apply the correct relationship types:
 - **One-to-One (1:1)**
 - **One-to-Many (1:M)**
 - **Many-to-Many (M:M)**
- Hint: Use **foreign keys** to create these relationships.

3. Create Tables with SQL

- Write CREATE TABLE statements for all entities.
- Choose appropriate **data types** (e.g., INT, VARCHAR, DATE, DECIMAL).
- Add **primary keys** and **foreign keys**.

4. Perform CRUD Operations

- **Create:** Insert at least **5 rows of sample data** into each table.
- **Read:** Use SELECT queries to fetch data.
- **Update:** Modify existing records using UPDATE.
- **Delete:** Remove specific records using DELETE.

5. Write Advanced Queries

- Filtering and sorting data.
- Modify data with UPDATE and DELETE.
- Create **User Defined Functions (UDFs)**.
- Perform **JOINS** and **Subqueries**.

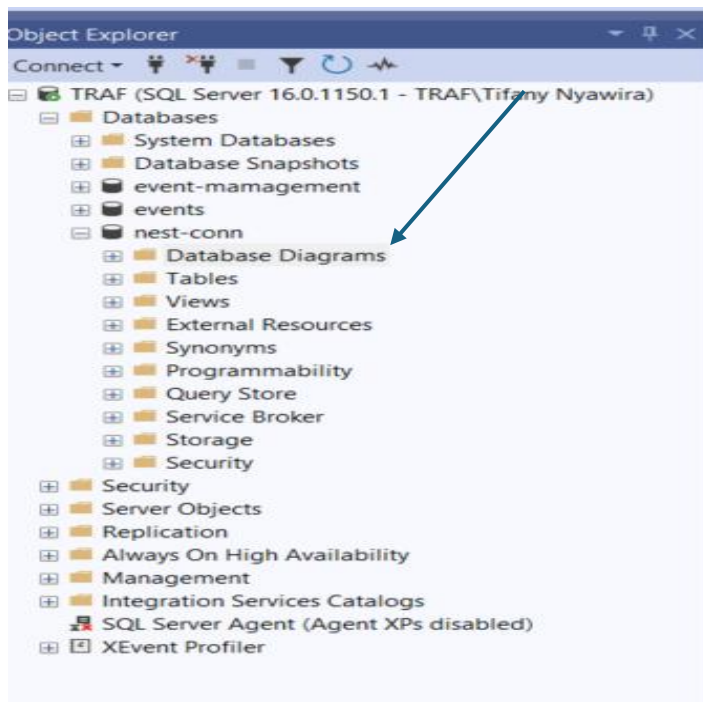
Car Rental Management System DB

Entities(tables):

1. Car • Car ID • Car Model • Manufacturer • Year • Color • Rental Rate • Availability	2. Customer • Customer ID • First Name • Last Name • Email • Phone Number • Address
3. Rental • Rental ID • Car ID • Customer ID • Rental Start Date • Rental End Date • Total Amount	4. Payment • Payment ID • Rental ID • Payment Date • Amount • Payment Method
5. Insurance • Insurance ID • Car ID • Insurance Provider • Policy Number • Start Date • End Date	6. Location • Location ID • Location Name • Address • Contact Number
7. Reservation • Reservation ID • Car ID • Customer ID • Reservation Date • Pickup Date • Return Date	8. Maintenance • Maintenance ID • Car ID • Maintenance Date • Description • Cost

Submission Guidelines:

- Save all your **SQL queries** in a .sql file.
- Create a Database Diagram in SSMS:
Right-click your diagram → Copy Diagram to Clipboard → Paste as an image.
- Ensure foreign keys are created so relationships appear in the SSMS diagram.



- Upload your .sql file and diagram image to **GitHub**.
- Submit your **individual work** as GitHub link to: https://teach2give-my.sharepoint.com/:x:/p/tiffany_mwaniki/EQ764vqGq4tGtJkqrrHf394BYHtiR0UYAzdobRwEoi06uQ?e=xJLUJw